

The Wrong Unit of Uncertainty: Simultaneous Conformal Bands for Repeated Cross-National Surveys

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Abstract

Comparative research routinely reads a single wave-pair mean contrast as a persistent, distribution-wide, country-level trend—attaching uncertainty to the wrong unit and over-detecting change. We give a deployable instrument: a finite-sample simultaneous band over a country’s whole attitude *trajectory*, with the country as the exchangeable unit and a hierarchy of claims whose top rung is the honest object. The band is a clustered conformal band exact at any K , inside an adaptive selector whose validity is selection-free—its fallback branches are nested, so at survey scale the deployed guarantee is exact under *any* selection rule—and which corrects for survey design only at the many-unit scale where that correction is provably reliable. In the ESS (rounds 9–11), net decline in parliamentary trust is certified in six of thirty countries and persistent decline in one (Greece, on a single wave pair)—not the twenty flagged marginally. On the World Values Survey (1981–2022), a marginal wave-pair reading over-counts the persistent phenomenon by 2.6–6.5 $\times$, and what survives concentrates in post-communist and Arab-Spring-aftermath states, not the consolidated West. A subtler second error—the calibration objects are themselves estimated from complex samples—we resolve by scope: the correction is non-identified without the design-noise law and, for the deployed procedure, unreachable at survey scale, so abstention is principled.

1 Introduction

Comparative politics increasingly treats a national public as a *distribution* rather than a mean. Trust in parliament, satisfaction with democracy, or support for the political system in a country and year is a whole cumulative distribution function over a response scale, and repeated cross-national surveys—the European Social Survey (ESS; Kaminska and Lynn, 2017), the World Values Survey, the AmericasBarometer (Cohen et al., 2021)—let us track how that distribution moves across waves and, increasingly, *transport* it: predict one country’s trajectory from the others, or flag which countries depart from a common pattern. Prediction bands and conformal methods are a natural tool here, promising finite-sample, distribution-free coverage.

Two wrong units. A claim and its calibration can each be attached to the wrong object. The *claim* “democratic trust is declining” is typically built from a single wave-pair mean contrast, then narrated as a persistent, distribution-wide, country-level trend. Requiring the whole distribution to shift, *simultaneously* across the entire trajectory, is a strictly stronger and more honest object, and the number of countries that qualify collapses between the two. Our instrument is a finite-sample simultaneous band over a country’s whole trajectory, with the country as the exchangeable unit, and an ordered hierarchy of claims (pairwise $\rightarrow$ any-pair $\rightarrow$ net $\rightarrow$ persistent country-wide $\rightarrow$ Bonferroni-across-countries) whose top rung is the object casual readings implicitly assert. This multiplicity-honest core needs no survey design at all and applies to any repeated distributional claim.

The second, subtler wrong unit is a layer of uncertainty these exercises silently omit. Conformal inference is usually calibrated on *observed outcomes*. Repeated cross-national surveys are different: the calibration objects—the other countries’ attitude distributions—are themselves *not observed*. Each is estimated from a complex probability sample with weights, clustering (primary sampling units), and stratification, and therefore carries its own sampling error. Standard practice plugs these survey estimates into the calibration step as

if they were the truth, so the transport band, honest about variation *across* countries, stays blind to the fact that each “country” is a noisy estimate of a latent population function.

The two errors are genuinely distinct. Writing $F_{c,r}(t)$ for country c ’s latent attitude CDF at threshold t in wave r , and $\tilde{F}_{c,r}(t)$ for its complex-survey estimate,

$$\tilde{F}_{c,r}(t) - \mu_r(t) = \underbrace{(F_{c,r}(t) - \mu_r(t))}_{\text{transport error}} + \underbrace{(\tilde{F}_{c,r}(t) - F_{c,r}(t))}_{\text{design error}}, \quad (1)$$

where μ_r is the transport center. A conformal band calibrated on the observed $\tilde{F}$ scores treats the sum as if it were the transport error alone: when the design error is non-negligible the band is systematically too wide, yet any correction that subtracts the survey noise can make it too narrow, because the correction is itself estimated from a small number of countries.

Contributions. We make four. *First, a diagnosis and a portable instrument.* We separate the two wrong units—the claim unit (a wave-pair contrast standing in for a persistent distribution-wide trajectory) and the calibration unit (estimated distributions treated as latent, the omitted design-error layer of (1))—and give a finite-sample simultaneous band over a country’s whole trajectory, with a multiplicity-honest guarantee hierarchy that corrects the first. *Second, a provably-valid deployed procedure with an unconditional guarantee.* The band we recommend is a clustered “population” conformal band (PCB), finite-sample exact at *any* K (Theorem 3), inside an adaptive selector whose validity does not depend on how it selects: the PCB and conservative branches are *nested*, and selection among nested bands anchored at an exact band is validity-free, so the four target-blind gates are an efficiency device, not a validity assumption (Theorem 5). The design-error-correcting branch carries its own miss budget, and only at the many-unit scale where it can activate at all; at cross-national-survey K the deployed guarantee is exact $1 - \alpha$, full stop. We validate the frozen procedure in three disclosed layers, ending with a fresh sealed run on six data-generating-process families its design had never seen (Section 6). *Third, two named reanalyses* (Section 8) where the

instrument changes the substantive picture. For European trust (ESS 2018–2022), marginal wave-by-wave estimates suggest widespread decline, but a country-wide, design-aware simultaneous band certifies a *persistent distribution-wide* decline in only one country (Greece), not the twenty flagged marginally. Globally, we reanalyze the Foa–Mounk “democratic deconsolidation” thesis on the World Values Survey (1981–2022) and find its marginal evidence over-counts persistent, distribution-wide deconsolidation by $2.6\text{--}6.5\times$ —real in a specific minority, not the broad erosion the marginal reading implies. Aggregating the per-item certifications turns the correction into a discovery: the *certified core* of multi-item deconsolidation (thirteen countries) is concentrated in post-communist and Arab-Spring-aftermath states, with the consolidated West entering only through rising openness to strongman and army rule (Finland, Switzerland)—the deconsolidation that survives the honest bar lives largely outside the mature democracies the thesis was formulated about. *Fourth, the scope condition that justifies this deployment.* We delineate exactly when a design-aware correction is available and when it is not, so that abstention is a principled choice rather than a concession. Without a model of the design-noise law the latent transport-error distribution is non-identified—a deconvolution problem (Theorem 1); and even *with* that information the correction is *unreachable* at cross-national scale for the deployed procedure at its frozen constants, for two quantitative reasons: the measured design-to-total noise ratio ρ stays well below the level at which deconvolution helps (the estimates are downward-biased floors, and still severalfold short), and the deployed reliability diagnostic obeys the algorithmic floor $D \geq \sqrt{2/(K-1)}$ —an identity of the frozen procedure, holding for every possible dataset—that needs $K \geq 94$ exchangeable populations. The calibration-free content of the second barrier is the variance-estimation floor itself; a differently calibrated correction could move the $K \geq 94$ threshold, though not below the floor’s own arithmetic. On the two largest such surveys these barriers bind in *opposite* ways—the World Values Survey clears the size barrier ($K \approx 100$) but carries only weights-based, negligible design noise, while the ESS has genuine clustered design noise but far too few countries—so neither clears both. This is why

the abstaining reduction above, not a design-aware correction, is the right object at survey scale. The empirical payoff is thus the two reanalyses; the design-aware layer is a diagnostic and scope result that, on real survey data, certifies the simple band rather than firing—so the deconvolution is not something the reader should expect to activate empirically.

The remainder situates the contribution among the conformal and political-trust literatures (Section 2), sets up the problem and impossibility result (Section 3), builds the safe adaptive method and its theory (Sections 4–5), validates it in simulation (Section 6), characterizes the real-data regime (Section 7), reanalyzes political trust (Section 8), and discusses scope (Section 9).

2 Related work

Our work sits at the intersection of four literatures. First, **distribution-free conformal prediction** gives finite-sample coverage without modeling assumptions (Vovk et al., 2005; Lei et al., 2018). Because political data rarely satisfy exchangeability, we build on its relaxations: weighted conformal prediction for covariate shift (Tibshirani et al., 2019) and nonexchangeable conformal prediction, which bounds the coverage gap by the total variation between exchanged distributions (Barber et al., 2023). Gibbs and Candès (2021); Gibbs et al. (2024) adapt conformal sets to distribution shift and interpolate between marginal and conditional validity. Our “adaptive” selector differs in kind: it adapts not to temporal shift but to the *certified magnitude of survey design noise*, reducing to a fixed clustered method when that noise is negligible. Within the discipline, Randahl et al. (2026) bring conformal prediction to *Political Analysis* for the first time, calibrating bin-conditional intervals around a point forecast of conflict fatalities; our object is different in kind—finite-sample *simultaneous* trajectory bands with the country as the exchangeable unit and estimated calibration distributions, rather than bin-conditional intervals on a single prediction.

Second, we treat the **population (country) as the exchangeable unit**, following

hierarchical conformal prediction for two-layer models (Dunn et al., 2023) and distribution-free inference with hierarchical data (Lee et al., 2023). The same finite-sample-exact clustered construction is developed, for a different (non-survey) domain, in a companion paper by the present authors (reference omitted for anonymous review). These methods transport guarantees to an unseen cluster—our cross-national setting—but treat each cluster’s data as cleanly sampled. Our departure is that the calibration objects are not observed: each country’s attitude distribution is *estimated* under a complex survey design, so design-based sampling variance (Binder, 1983; Lumley, 2010) contaminates the very scores conformal calibration relies upon.

Third, the closest methodological neighbors handle **noisy or estimated calibration**. Wieczorek (2023) develops design-based conformal prediction under complex sampling, but design enters through test-point weighting rather than as noise in transported objects. Einbinder et al. (2024) and Sesia et al. (2025) correct conformal calibration under label noise, the latter by deconvolving a *known* noise model. Our central theoretical result is that this identification premise generally fails for survey design noise: recovering the latent transport-error law is a deconvolution problem (Fan, 1991) that is non-identified without a design-noise law, so no observed-score band is simultaneously valid and strictly narrower than the conservative plug-in. And even when a noise model *is* supplied, we show the deconvolution’s reliability is itself bounded at a finite number of calibration units by an algorithmic floor (Proposition 1)—a finite- K obstruction distinct from the asymptotic identifiability the classical deconvolution literature studies.

Fourth, our target is a **functional object**—a CDF over thresholds tracked across rounds—placing us near simultaneous functional conformal bands (Diquigiovanni et al., 2021). This is where we part from generic multiplicity control. Procedures like FDR or Bonferroni (Benjamini and Hochberg, 1995) adjust a collection of tests after the fact; our guarantee is instead carried by the geometry of a single finite-sample band—a supremum over wave-pairs and thresholds with the country as the exchangeable unit—so the “persistent, distribution-

wide” object is *defined and covered directly*, not recovered by correcting marginal p -values. The distinction is one of estimand, not merely error rate, and partial-pooling cautions about reading marginal country trends (Gelman et al., 2012) motivate but do not deliver the stronger claim. Substantively, our two reanalyses supply the formal instrument a largely qualitative skepticism has lacked, engaging debates over whether European political trust is in broad decline (Norris, 2017; Voeten, 2017; Wuttke et al., 2022; Valgarðsson et al., 2025) and whether publics are “deconsolidating” (Foa and Mounk, 2016, 2017). The closest substantive competitor is Claassen (2019, 2020), whose latent-variable panels of democratic support—published in this journal—already model measurement and sampling uncertainty in cross-national attitude trends. The difference is the type of guarantee and the object: Claassen-style smoothing delivers model-based credible intervals for a latent mood under a dynamic measurement model, pooling across items and years; our bands deliver finite-sample, model-free coverage for a distribution-level object at a stated claim rung, at the price of wider intervals and no pooling. The two are complements—a certification that survives our rung hierarchy and falls inside a Claassen-style interval is doubly supported—and a same-data comparison is a natural companion exercise. The geography of our certified core also intersects the post-communist disillusionment literature (Mishler and Rose, 1997; Inglehart and Catterberg, 2003), which anticipated qualitatively the decline-from-euphoric-baselines pattern our per-country certification formalizes.

3 Setup and the two targets

The object we deploy is the exact simultaneous band of Theorem 3; the impossibility below (Theorem 1) is a scope boundary delimiting when a design-aware correction to that band is even available.

Objects. We observe K source countries, indexed $c = 1, \dots, K$, treated as exchangeable units (the “population” or cluster is the exchangeable object, not the individual respondent).

Each country has a latent attitude trajectory $F_{c,r}(t)$, a CDF over a fixed grid of thresholds $t \in \{t_1, \dots, t_T\}$ and survey rounds r . We do not see $F_{c,r}$; we see a complex-survey estimate $\tilde{F}_{c,r}(t) = F_{c,r}(t) + S_{c,r}(t)$, where $S_{c,r}$ is design (sampling) error with a design variance $v_{c,r}(t)^2$ that a stratified-PSU / replicate-weight bootstrap can estimate. Define the transport score of a country as its worst-case standardized deviation from the transport center,

$$R_c = \max_{r,t} \frac{|F_{c,r}(t) - \mu_r(t)|}{\sigma(t)}, \quad \tilde{R}_c = R_c + \xi_c,$$

where $\tilde{R}_c$ is the same quantity computed from the observed $\tilde{F}$ and ξ_c is the induced design contamination. (Because R is a supremum, the induced ξ_c is not exactly mean-zero or independent of R_c —taking a max over noisy coordinates biases the score upward; the additive, centered score model here is a working approximation at the score scale, made exact at the curve scale by the scale-family assumption of Theorem 4, and the direction of its failure—upward-biased observed scores—is conservative for the plug-in band.) The scalar that governs the whole problem is the design-to-*total* standard-deviation ratio

$$\rho = \frac{\text{SD}(\xi)}{\text{SD}(\tilde{R})} = \sqrt{\frac{v^2}{s_R^2 + v^2}} \in [0, 1),$$

with $\rho \rightarrow 0$ the design-noise-negligible regime and $\rho \rightarrow 1$ the regime in which survey noise dominates the between-country spread. (The generative dial in the simulations is the unbounded design/transport ratio $\text{SD}(\xi)/\text{SD}(R)$; all reported ρ values, gates, and the $\sqrt{1 - \rho^2}$ width law use the bounded design/total definition above.)

Target and validity. We want a simultaneous band $\hat{B}(r, t)$ that covers a *new* country’s *latent* trajectory, $\Pr(F_{\text{new},r}(t) \in \hat{B}(r, t) \forall r, t) \geq 1 - \alpha$. Two facts frame everything. Ordinary clustered conformal on the observed scores is exact for the *survey-estimate* target $\tilde{F}_{\text{new}}$ (it only needs exchangeability of countries); its coverage of the *latent* target differs by an explicit gap that vanishes with the design-noise scale (Theorem 2(b))—symmetry of the design law

alone is *not* enough for conservativeness, so we bound the gap rather than assume its sign). Meanwhile its width is inflated by the design noise, by a factor that grows with ρ . Correcting the width requires estimating and removing the design contribution—a deconvolution.

Country exchangeability, stated honestly. The transport band’s identifying assumption is exchangeability of countries, and we state what that means: a superpopulation postulate that the participating countries’ trajectory laws are symmetric draws—over a set that is not sampled, whose members enter and exit rounds non-randomly, and whose within-country designs change across rounds. It is a strong idealization (and not comparable to, much less weaker than, the i.i.d.-respondents assumption, which operates at a different layer inside an inspectable design). Three things temper it. First, the headline certifications of Section 8 do not use it: they are within-country design-bootstrap statements in which the country effect cancels. Second, when exchangeability holds only approximately, the nonexchangeable conformal bound of Barber et al. (2023) controls the transport band’s coverage gap by a weighted total-variation distance between country laws; we do not observe that distance, so we treat the bound as qualitative and flag a leave-one-region-out calibration audit for the archived replication rather than claiming graceful degradation as fact. Third, the conservative (Bonferroni-across-countries) rung protects against the family-wise multiplicity of screening many countries—a distinct safeguard that does not repair non-exchangeability.

The impossibility. The central obstruction is that the deconvolution is not identified from the contaminated scores alone.

Theorem 1 (Non-identification and necessity of design information). *Let the observed score be $\tilde{R} = R + \xi$ with $R \perp \xi$. (i) The law of R is not identified from the law of $\tilde{R}$ without restrictions on the law of ξ : there exist distinct pairs $(\mathcal{L}_R, \mathcal{L}_\xi) \neq (\mathcal{L}'_R, \mathcal{L}'_\xi)$ with $\mathcal{L}_R * \mathcal{L}_\xi = \mathcal{L}_{R'} * \mathcal{L}_{\xi'}$. (ii) Consequently, among non-randomized bands that are measurable functions of the observed scores only, any band valid for the latent target on a class of design laws containing $\xi \equiv 0$ must be at least as wide as the plug-in band whose radius is the $(1 - \alpha)$*

conformal order statistic of $|\tilde{R}|$, up to the conformal discreteness slack. No observed-score band can beat the plug-in by more than that slack without shrinking the class of admissible noise laws.

The proof (Appendix A, with a numerical contract test) is a convolution-invertibility argument for (i) and a conformal converse for (ii). The reading is constructive: a band both honest and narrower than the plug-in must *supply* the design-noise law, which complex surveys do provide—through replicate weights or a design bootstrap, exactly what Section 4 uses—so this input is not optional.

Remark 1 (Beyond surveys). Nothing in Theorem 1 is specific to surveys: it holds whenever a conformal procedure is calibrated on *estimated* objects, so that the nonconformity score is a noisy version $\tilde{R} = R + \xi$ of a latent score—model-based small-area (MRP) outputs, learned or imputed features, meta-analytic effects. The same two-part consequence follows: design/measurement information is the identifying ingredient, and its reliability at a finite number of calibration units is itself bounded (Proposition 1). The barrier is confirmed outside surveys on a model-based small-area calibration (Supplementary Material).

4 A nominal-safe adaptive method

Theorem 1 says the design bootstrap is the identifying input. The method turns that input into a band by choosing, target-blind, among three constructions and by never using the aggressive one unless it is provably reliable at the available number of countries.

Three branches. For calibration error curves $\{\tilde{E}_c\}_{c=1}^K$ over threshold grid t , with per-country design SD $\hat{v}_c(t)$ from the stratified-PSU bootstrap: (i) *Clustered PCB* takes the conformal quantile of the *unstudentized* scores $\max_t |\tilde{E}_c(t)|$ (the U0 construction of Theorem 3); it is finite-sample exact for the survey-estimate target at any K , and its latent-target gap is the explicit two-term bound of Theorem 2(b), vanishing with the design-noise scale.

(ii) *Deconvolution* studentizes by the finite- K -safe target scale $\widehat{s}_T^2(t) = \max(s_{\text{plug}}^2(t) - [\widehat{v}^2(t) - z \text{SE}]_+, \text{floor})$, which subtracts a *lower* confidence bound on the design variance so the correction cannot over-shrink; its width scales as $\sqrt{1 - \rho^2}$ relative to PCB. (iii) *Conservative* inflates each unstudentized score by the design SD, $\max_t(|\widetilde{E}_c(t)| + z \widehat{v}_c(t))$ — a score that dominates the PCB score pointwise, so the conservative band *contains* the PCB band by construction. This nesting is load-bearing: selection between nested bands anchored at an exact band is validity-free (Theorem 5), so the selector needs no conditional-coverage argument.

The three questions. The naive selector asks only *can we deconvolve* and *do we need to*. The finite- K failure of deconvolution (Section 6) comes from a missing third question: *can we do it safely at this K ?* The safe selector adds it explicitly.

Safe selector (Algorithm 1). Let $\widehat{\rho}_{\text{LCB}}$ be a conservative lower bound on ρ , $D = \max_t \text{SE}(\widehat{s}_T^2)/\widehat{s}_T^2$ a reliability diagnostic, and $\widehat{\delta}_{\text{UCB}}(D)$ a frozen upper bound on the finite- K coverage remainder. Deconvolution is used *iff all four gates pass*: (A) $\widehat{\rho}_{\text{LCB}} > \rho_0$ (design noise materially large); (B) $\widehat{\delta}_{\text{UCB}}(D) \leq \delta_{\text{max}}$ (remainder within budget); (C) $\widehat{\text{Gain}} - \Delta_g > g_{\text{min}}$ (beats conservative by a margin); (D) stability. Otherwise the band is PCB when $\widehat{\rho}_{\text{LCB}} \leq \rho_0$ and the conservative envelope when design noise is real but information insufficient. Every gate is a function of the calibration set only, so the selector is target-blind — though, by the nested-band argument of Theorem 5, validity would survive even an adversarial selection between the two anchor branches. The deconvolution branch, which is *not* nested, is charged its own miss budget $\alpha_{\text{dec}} = \min\{\max(0.1\alpha, 3/(K+1)), \alpha/2\}$ (the anchors run at $\alpha - \alpha_{\text{dec}}$), and only in the $K \geq 94$ regime where its gate can open at all; at the K of every certification sample in this paper the anchors keep the full α and the deployed guarantee is exact. All constants are frozen before validation (Section 5, Appendix B); when deconvolution is used the reported guarantee is the observable $1 - \alpha - \widehat{\delta}_{\text{UCB}}(D)$.

Algorithm 1: Nominal-safe adaptive design-aware band

Input: calibration curves $\{\tilde{R}_c\}$ (unit = country), design replicates $\{\hat{v}_c\}$, target curve $\hat{F}_{\text{new}}$, level α ; frozen $\rho_0, \delta_{\max}, g_{\min}, \Delta_g$ and the K -only budget rule $\alpha_{\text{dec}}(K)$ set $(\alpha_{\text{anchor}}, \alpha_{\text{dec}})$ by the frozen budget rule (full α to the anchors when gate B is infeasible at this K);
compute diagnostics $s, \hat{s}_T, \hat{\rho}_{\text{LCB}}, D, \hat{\delta}_{\text{UCB}}(D)$; the nested unstudentized anchor radii $r_{\text{PCB}}, r_{\text{con}}$ at level α_{anchor} ; and r_{dec} at level α_{dec} ;
if $\hat{\rho}_{\text{LCB}} \leq \rho_0$ **then**
| branch $\leftarrow$ PCB;
else if $\hat{\delta}_{\text{UCB}}(D) \leq \delta_{\max}$ **and** *stable* **and** $\widehat{\text{Gain}} - \Delta_g > g_{\min}$ **then**
| branch $\leftarrow$ deconvolution;
else
| branch $\leftarrow$ conservative;
return $\hat{F}_{\text{new}} \pm r_{\text{branch}}$, branch, and diagnostics;

Deployment. The method ships as one call, `dapcb(cal_errors, v_cal, center, alpha)`, in an installable Python package (module `pcb`) provided in the replication package and to be archived with a version tag on the Dataverse, returning the band together with the selected branch, $\hat{\rho}, \hat{\rho}_{\text{LCB}}, D, \hat{\delta}_{\text{UCB}}$, the coverage level, and the fallback reason—so a user sees *why* the band was built as it was and reads off the finite- K guarantee it carries.

5 Theory

We state validity for the oracle, for the estimated-law procedure, and for the deployed safe selector, then the efficiency it buys. Full proofs are in Appendix A; each theorem has a matching machine-checked contract test in the replication package.

Theorem 2 (Oracle validity). *Suppose the countries are exchangeable and the design law is known. (a) The clustered band is finite-sample valid for the survey-estimate target: $\Pr(\tilde{F}_{\text{new}} \in \hat{B}) \geq 1 - \alpha$, using exchangeability only. (b) For the latent target the same band satisfies, for every $u > 0$, $\Pr(F_{\text{new}} \in \hat{B}) \geq 1 - \alpha - \Pr(\xi < -u) - \Pr(\tilde{R} \in (\hat{q} - u, \hat{q}])$: the gap between the two targets is a noise-tail term plus a local anti-concentration term, both vanishing with the design-noise scale (of order $\rho^{2/3}$ under variance-only tail bounds and ρ up to logarithms under*

sub-Gaussian design noise; Appendix A states the conditions, including where independence across countries is used), and zero by inspection when the design noise is zero. Symmetric design noise alone does not make the band conservative for the latent target — a bimodal latent score law furnishes a counterexample (Supplementary Material, Remark on the latent-target bridge) — which is why we report the observed ρ that makes the gap negligible rather than assuming conservativeness. (c) The deconvolution band with the true target scale is valid for the latent target—finite-sample exact under the scale-family assumption (A1) of Theorem 4—with width a factor $\sqrt{1 - \rho^2} + o(1)$ of the clustered band.

Theorem 3 (Exact finite- K validity of the deployed base band). *Let the base band use the unstudentized cluster score $R_c = \max_t |E_c(t)|$ (no data-dependent modulation), with a center satisfying the symmetric-construction condition: fixed, split-fold, per-unit from that unit’s own data (the deployed own-previous-round centers), or a symmetric function of all $K+1$ curves (Appendix A states the four cases). If the countries are exchangeable, then for every K the band $\widehat{B} = \text{center} \pm \widehat{q}$, with $\widehat{q}$ the $[(1 - \alpha)(K+1)]$ -th order statistic of $\{R_c\}$, satisfies $\Pr(F_{\text{new}} \in \widehat{B}) \geq [(1 - \alpha)(K+1)]/(K+1) \geq 1 - \alpha$, distribution-free. Two asymmetries are excluded by construction and quantified when present: dividing R_c by an in-sample modulation $\widehat{s}(t) = \text{SD}_c E_c(t)$ —which the unobserved target does not share—induces an $O(g/K)$ self-inclusion deficit, and centering calibration scores on a grand mean that contains their own unit while the unsurveyed target sits outside induces an anti-conservative $O(1/K)$ bias, removed by leave-one-out centers (Appendix A). We therefore deploy the unstudentized band with symmetric centers; on the near-homoskedastic real curves the width cost of dropping the modulation is $\leq 5\%$.*

Theorem 4 (Estimated-law validity). *Assume the observed and latent score curves sit in a common scale family indexed by the design variance (A1), the sup-score law satisfies an anti-concentration bound near its upper quantile (A2), and fourth moments and an unbiased size- B design bootstrap exist (A3) — Appendix A states all three precisely. Then (i) with oracle scales the deconvolution band is finite-sample exact for the latent target at*

every K , and (ii) with estimated scales it satisfies $\Pr(F_{new} \in \widehat{B}) \geq 1 - \alpha - \varepsilon_{K,B}$ with $\varepsilon_{K,B} = O(\sqrt{\log(KT)/\min(K,B)})$, proved by a sandwich and concentration argument with explicit constants. The finite- K -safe scale $\widehat{s}_T^2$ (Section 4) subtracts a lower confidence bound on the design variance, making the scale-error term one-sided (wide, not narrow) outside an α_2 -probability event. Some shape restriction is not removable—the bimodal counterexample of Theorem 2(b) defeats any latent-target band without one—and (A1) is one sufficient choice, strong enough to make the distributional matching exact; alternatives (e.g. smoothness conditions under a known noise law, as in the deconvolution literature) trade finite-sample exactness for weaker assumptions.

Theorem 5 (Safe-adaptive finite- K validity—the deployed pipeline). *Deploy Algorithm 1 with unstudentized PCB and conservative branches (scores $\max_t |E_c(t)|$ and $\max_t (|E_c(t)| + z\widehat{v}_c(t))$, both at level α_{anchor}) and the deconvolution branch at its own budget α_{dec} , where $(\alpha_{anchor}, \alpha_{dec}) = (\alpha, 0)$ when the reliability gate is infeasible at the observed K (Proposition 1: $K < 94$, which covers every certification sample analyzed in this paper, including the WVS ≥ 2 -wave sets at $K = 59$ –77) and otherwise $\alpha_{dec} = \min\{\max(0.1\alpha, 3/(K+1)), \alpha/2\}$ with $\alpha_{anchor} = \alpha - \alpha_{dec}$, a frozen function of K alone. Then, finite-sample and with no conditions on the selection rule,*

$$\Pr(F_{new,r}(t) \in \widehat{B}(r,t) \ \forall r,t) \geq 1 - \alpha - \varepsilon_{K,B} \mathbf{1}\{K \geq 94\},$$

where $\varepsilon_{K,B}$ is Theorem 4’s estimated-law remainder, attached to the deconvolution branch only—the $K \geq 94$ clause therefore inherits that theorem’s assumptions (A1)–(A3), while the $K < 94$ clause needs exchangeability only—and reported through the observable diagnostic $\widehat{\delta}_{UCB}(D)$ when that branch fires.

The proof (Appendix A) uses no conditional-on-selection step. Its two ingredients are a nested-band lemma—the conservative score dominates the PCB score pointwise, so the two bands are nested, and *any* selection between nested bands anchored at an exact band is

validity-free—and a union bound that charges the non-nested deconvolution branch its own α_{dec} miss budget, but only in the $K \geq 94$ regime where Proposition 1 shows its gate can open at all. The gates of Algorithm 1 therefore carry no validity burden; they are an efficiency device choosing among individually guaranteed constructions. At survey scale the deployed guarantee is exact $1 - \alpha$ with no remainder — the guarantee a user actually deploys — and the only remainder anywhere in the pipeline is confined to the many-unit regime where deconvolution can activate.

Efficiency. Three statements bound the price and payoff, with unequal epistemic status stated plainly. (*AW-1, proved*) in the low- ρ regime the deconvolution/PCB width ratio is $1 - \frac{1}{2}\rho^2 + O(\rho^4)$, matching the observed reduction to the decimal. (*AW-2, partly by construction, partly conditional*) the conservative branch is at least as wide as PCB—under the deployed unstudentized construction this dominance holds pointwise by construction, making the two branches nested—while its domination *by* the Bonferroni envelope rests on a stated first-order expansion and is not separately proved. (*AW-3, proved*) selection cannot inflate miscoverage, via the nested-band lemma and union bound of Theorem 5 rather than any conditional-coverage claim. The excess-width oracle inequality (AW-4, open) and the $O(g/K)$ modulation self-inclusion deficit—empirically fitted, with the scalar-invariance lemma isolating what it measures—are stated in Appendix A.

6 Simulation and a design-preserving stress test

Simulation establishes two facts: that attaching the band to the wrong unit collapses coverage (Section 6.1), and that the safe selector meets its guarantees on data it has never seen. For the latter we develop the selector on one simulation grid and then confirm the *frozen* rule once on an independent design. The separation is deliberate: the development grid may set the gate constants, but it may not be reused as validation performance.

Table 1: Whole-trajectory coverage (4000 reps, nominal 90%) as the trajectory length L grows. A band calibrated per point or per round covers each of its own units at the nominal rate, but the probability that it covers the *entire* trajectory decays; the country-trajectory band—the correct unit—holds nominal at every L .

L (rounds)	marginal (per point)	per-round	trajectory (ours)
2	38.2%	81.7%	90.0%
4	16.1%	68.2%	90.2%
6	7.1%	58.2%	90.3%
8	3.5%	49.8%	90.9%

6.1 The wrong unit collapses coverage

The most basic claim deserves a direct test: a band attached to the wrong unit covers the object of interest far below nominal. On a ground-truth simulation ($K = 30$ exchangeable countries, $T = 6$ thresholds, error tensors correlated across rounds and thresholds; Table 1), we ask each band for *trajectory* coverage—the whole country curve-path covered at once. All three bands are unstudentized and finite-sample constructed, so the only thing that varies is the unit of the score.

The per-round band, marginally valid at the round level, covers the whole trajectory only 0.82 at $L = 2$ and 0.50 at $L = 8$; the marginal band collapses faster still. The country-trajectory band holds ≈ 0.90 regardless of L , because the whole trajectory is a single exchangeable unit and its sup-score obeys the finite-sample order-statistic guarantee. This is the coverage face of the paper’s thesis: the same mechanism drives the certification-count collapse in Section 8, where asking for the honest object—not asking it more strictly—removes the over-detection. Pre-registered and reported as produced (`e28_wrong_unit_coverage`).

Severity: what each rung can detect. Coverage is only half of honesty; the other half is saying what magnitude of decline each rung would detect. A simulation calibrated to ESS-like design noise (threshold SE $v = 0.012$; sup- t certification at $\alpha = 0.10$ over a four-threshold core; replication experiment E32) gives the operating characteristics: for 80% power the *net* rung needs a per-pair decline of 0.02–0.03 CDF points over four pairs—realistic

secular erosion—while the *persistent* rung needs 0.06–0.08 per pair, roughly the scale of the Greek crisis collapse, and its power *falls* as the trajectory lengthens (the conjunction hardens while the per-pair signal does not accumulate). Two consequences discipline our substantive language. Non-certification at the persistent rung is weak evidence of absence—it is a high-severity, low-power object, informative when it *passes*; and for secular-decline claims the powered instrument is the net rung, which is why Section 8 co-headlines the net counts. Injecting known declines into the real microdata is the corresponding real-data check, flagged for the archived replication.

The development grid that motivates and calibrates the safety gate, and a controlled Gaussian illustration that isolates the finite- K over-shrinkage mechanism behind the $K=320$ -only activation seen below, are given in the Supplementary Material.

Validation, in three honestly-labeled layers. (i) *The sealed holdout* froze the selector and its constants (Appendix B), sealed a preregistration and seed manifest, and ran once on a grid disjoint from development in K ($\{25, 40, 80, 160, 320\}$), in ρ , and in data-generating process (ten families). Its confirmatory status was subsequently compromised in two disclosed ways: an audit found its scorer evaluated the S2 studentized band rather than the band the package deploys, and its findings motivated the switch to the U0 base band and the Theorem 5 architecture—so it functioned as development, and we label it so. (ii) *The corrected-scorer rerun* of the same 450-cell grid scores the deployed band verbatim (fresh seed; the original config file was not preserved, which we disclose). (iii) *A fresh seal*: because layers (i)–(ii) had both touched the final design, we sealed one further validation of the final pipeline—six new DGP families (lognormal country effects, AR(2) rounds, scale mixtures, extreme design-effect dispersion, threshold-correlated noise, long trajectories), new K and ρ grids, script hash recorded before first execution, run once, reported as produced including a first-seal generator bug we diagnosed and disclosed (replication experiment E33). Result: 120/120 cells floor-compatible, worst cell 0.878, and zero deconvolution activations

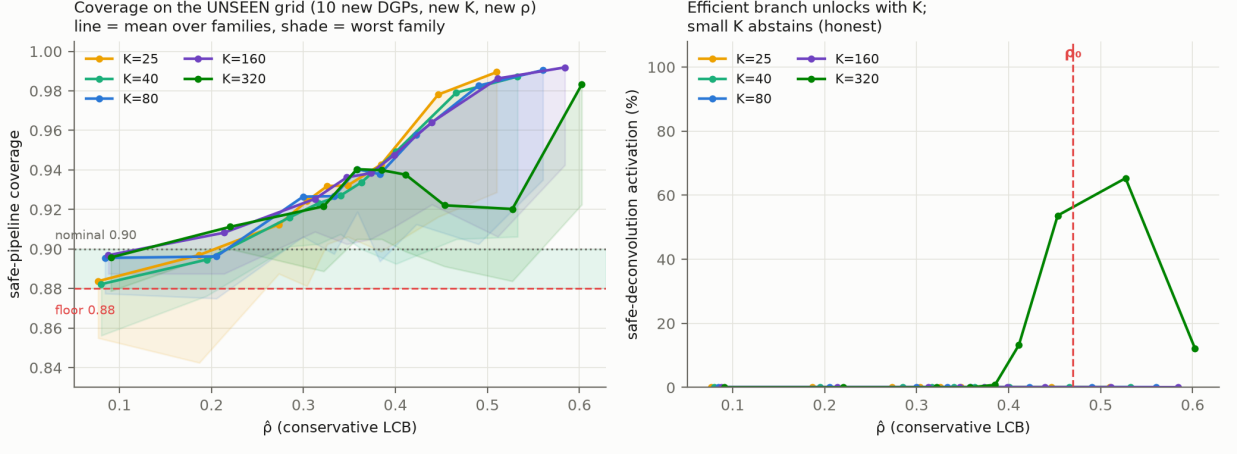


Figure 1: Independent confirmatory validation on the unseen grid (10 new DGP families, new K , new ρ). Left: deployed coverage vs. $\hat{\rho}$ per K (line = mean over families, shade = worst family) against the 0.88 floor and 0.90 nominal. Right: deconvolution activates only at $K=320$; small K abstains.

below $K = 94$, as the algorithmic floor requires. The numbers of all three layers are reported exactly as produced (Table 2, Figure 1).

The design-aware layer behaves as designed. In the sealed run all twelve cells below the 0.88 floor routed to base clustered PCB—the deconvolution branch never once rode into a floor violation. Under the corrected scorer the deployed pipeline passes every criterion outright: the selector reduces to PCB at low ρ (width ratio 1.005), activates deconvolution only at $K=320$ (never for $K \leq 160$; 24 cells, up to 99% of draws at the highest ρ), and there delivers a band $0.753\times$ the conservative width—wider than the sealed run’s 0.577 because the α_{dec} budget of Theorem 5 prices the branch’s estimated-law remainder honestly—with worst $K=320$ cell coverage 0.900.

The holdout exposed one base-method deficit, which we then resolved. The studentized base band (in-sample pooled modulation) undercovered at small K and low ρ under the hardest DGPs, because dividing each calibration score by a modulation estimated from the same clusters—which the unobserved target does not share—breaks the exchangeability the conformal guarantee needs. We therefore deploy the *unstudentized* clustered band as the base branch: with no data-dependent denominator the scores $R_c = \max_t |E_c(t)|$ are

Table 2: Holdout validation of the safe selector (450 cells, 800 reps each). The *sealed* run (upper rows) scored the studentized S2 base band, whose finite- K self-inclusion deficit it exposed — the finding that motivated deploying the *unstudentized* band instead. The *corrected-scorer rerun* (lower rows) scores the band the package actually returns (U0 anchors with the α -budgeted deconvolution branch of Theorem 5) on the same 450-cell grid with a fresh seed, disclosed because the originally sealed configuration file was not preserved in the [archive](#).

criterion	target	result	status
<i>Sealed run, S2 scorer (historical):</i>			
P3: design-aware never causes a sub-floor cell	sub-floor $\Rightarrow$ not deconv	all 12 sub-0.88 cells are PCB	pass
E2: deconvolution narrower than conservative	$\leq 0.90\times$	0.577	pass
P1: cells floor-compatible (95% MC)	all 450	442/450	diagnostic [†]
P2: worst-cell coverage	≥ 0.86	0.843	diagnostic [†]
<i>Corrected-scorer rerun, deployed U0/α-budget band:</i>			
P1: cells floor-compatible (95% MC)	all 450	450/450	pass
P2: worst-cell coverage	≥ 0.86	0.882	pass
E1: low- ρ width vs PCB	$\leq 1.05\times$	1.005	pass
E2: deconvolution narrower than conservative	$\leq 0.90\times$	0.753	pass
E3: deconvolution abstains at small K	rises with K	only $K=320$	pass

[†] Diagnostic of the studentized base band, not a deployed-band failure; these cells motivated the fix. The deployed unstudentized band is finite-sample exact at any K (Theorem 3); under the corrected scorer its worst holdout cell is 0.882 (1.65 MC-SE below the 0.90 nominal, i.e. binomial-noise compatible with exactness) and no cell fails the preregistered floor criterion.

exchangeable, and the order-statistic band is finite-sample exact at *any* K , $\Pr(\text{cover}) \geq [(1 - \alpha)(K+1)]/(K+1) \geq 1 - \alpha$, distribution-free (Theorem 3). A preregistered rerun on a fresh grid confirms the fix at a width cost of 1.02–1.05 $\times$ on the real ESS transport bands; the granular cell-by-cell results are in the Supplementary Material. The band the paper deploys is thus finite-sample valid at the sample sizes comparative politics actually has. A second, extreme-tail limitation of the Gaussian-calibrated remainder bound $\hat{\delta}_{\text{UCB}}$ —confined to 0.02% of draws and absorbed by the conservative branch—is documented in the Supplementary Material. We do not claim uniform nominal coverage; we claim—and the holdout confirms—that the design-aware correction is never the *cause* of undercoverage and is materially more efficient than conservative where it safely activates.

Design-preserving stress test, honestly bounded. A design-preserving subsampling of the real AmericasBarometer shows the deployed selector never fires even as PSUs are thinned toward the noise-dominated regime: pooled $\hat{\rho}$ rises to ≈ 0.23 and then *turns over*.

We flag the interpretation ourselves: under the protocol’s own $1/\sqrt{f}$ scaling the sweep was designed to cross ρ_0 at the deepest subsampling, and the turnover coincides with the point where ≈ 1 PSU remains per stratum—exactly where the m -of- m bootstrap’s variance estimate collapses. The experiment therefore demonstrates a failure mode of the $\hat{\rho}$ *diagnostic* under sparse-PSU designs (with the deployed band’s coverage remaining at or above nominal throughout), not a substantive saturation law; the regime characterization of Section 7 rests on the full-sample $\hat{\rho}$ values and the ESS subgroup scan, not on this sweep (Supplementary Material).

7 Evidence from the ESS and the AmericasBarometer

The regime characterization. On real cross-national data the honest band reduces to clustered conformal—and the reason is instructive. Across *both* surveys, *both* estimands (attitude levels and wave-to-wave change), and three outcomes (trust in parliament, satisfaction with democracy, system support), national cross-national inference is a uniformly *low design-noise* regime: the estimated $\hat{\rho} \leq 0.20$ (LAPOP, shown in Figure 2; the ESS subgroup scan reaches at most 0.29, reported below), comfortably below the $\rho_0 = 0.47$ cutoff, because between-country variation dwarfs within-survey design noise. Two directional caveats attach to every $\hat{\rho}$ we report: the PSU bootstrap resamples m -of- m without the Rao–Wu–Yue rescaling (Rao et al., 1992), and single-PSU strata contribute no variance, so design variance—hence $\hat{\rho}$ —is biased *downward* by an unquantified amount; the qualitative regime conclusion survives because the observed gap to ρ_0 is severalfold, but the numbers are floors, not measurements. Consequently the safe selector reduces to clustered PCB on *all* real data—not by assumption but by its own low- ρ gate. The elaborate design-aware machinery is, on real cross-national attitude data, correctly inactive.

This is a substantive finding: the impossibility of Section 3 bites in principle, but the deconvolution it necessitates is never triggered at survey scale, so honest inference here *is*

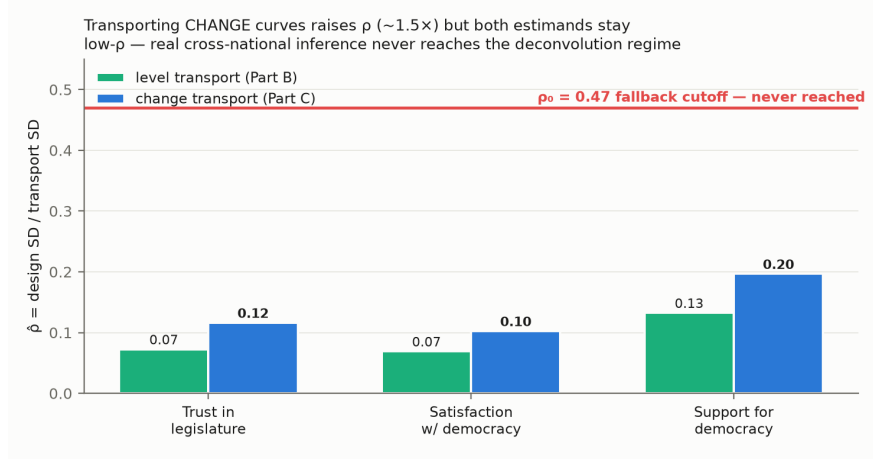


Figure 2: Estimated design-to-total SD ratio $\hat{\rho}$ for the AmericasBarometer, levels and wave-to-wave change. Every country-outcome sits well below the $\rho_0 = 0.47$ cutoff: national cross-national attitude inference is a uniformly low design-noise regime, so the safe selector reduces to clustered PCB.

clustered conformal on the observed scores.

Why the deconvolution is unreachable, precisely. The inactivity is not an accident of one dataset; it is forced by two structural barriers, and we can say how far each real survey is from clearing them.

Proposition 1 (Survey-scale unreachability). *The deployed selector activates the deconvolution branch only if both a need gate $\hat{\rho}_{\text{LCB}} > \rho_0$ and a reliability gate $D \leq \tau_D$ hold, where $D = \max_t \widehat{\text{SE}}(\hat{s}_T^2)/\hat{s}_T^2$. Two facts bound these, both algorithmic identities of the frozen procedure—they hold for every possible dataset, not on average (Appendix A separates this deterministic statement from its kurtosis-dependent statistical interpretation). (a) $\rho = \sqrt{v^2}/s_{\text{plug}} \in [0, 1)$ with $s_{\text{plug}}^2 = s_R^2 + \overline{v^2}$, so the need gate can fire only when the design noise is an appreciable fraction of the between-population signal s_R . (b) Since $\hat{s}_T \leq s_{\text{plug}}$ by construction, the reliability obeys $D \geq \sqrt{2/(K-1)}$; hence the reliability gate requires $K \geq 1 + 2/\tau_D^2$, i.e. $K \geq 94$ at the frozen $\tau_D = 0.147$, exchangeable populations.*

The two requirements pull apart on real surveys (Figure 3). On the ESS, narrowing to the noisiest subpopulations (youth age bands) raises $\hat{\rho}$ from 0.08 to at most 0.29—still far

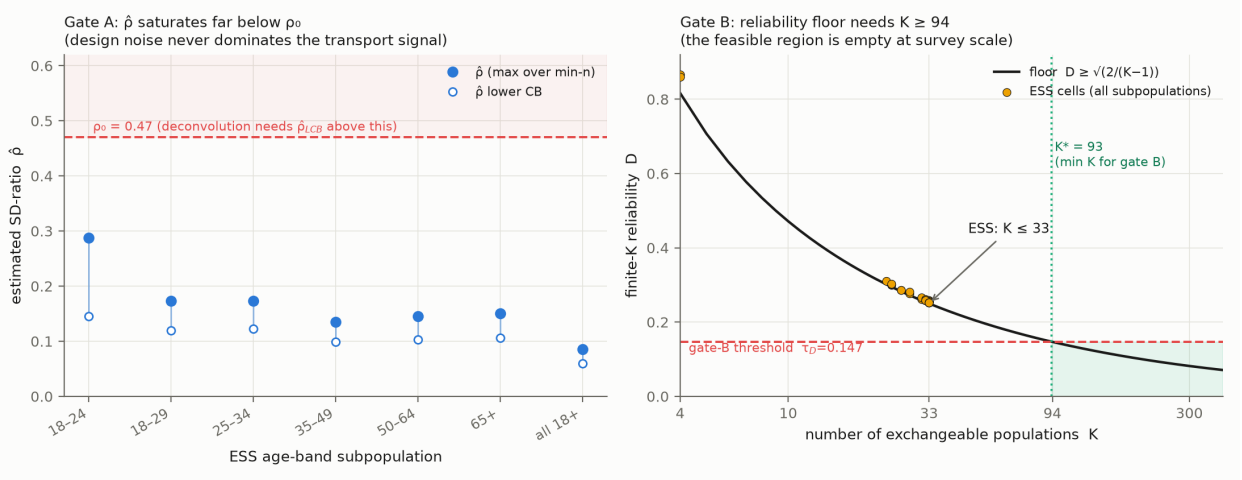


Figure 3: The two barriers, measured on real data. *Left*: across ESS age-band subpopulations $\hat{\rho}$ stays far below $\rho_0 = 0.47$ (max 0.29; the estimates are downward-biased floors, Section text). *Right*: the finite- K reliability D sits on its algorithmic floor $\sqrt{2/(K-1)}$ (deterministic for the frozen diagnostic; Proposition 1a); passing the reliability gate needs $K \geq 94$, which cross-national samples reach only in the WVS full item-coverage sets ($K \leq 105$)—where the design noise is too small for the need gate—and never in the certification samples ($K \leq 33$ ESS, 59–77 WVS).

below ρ_0 —while $K \leq 33$ leaves $D \geq 0.25$, so *neither* gate opens (a scan of 42 subpopulation cells activates deconvolution in none). The World Values Survey is the mirror image, with one denominator made explicit: the gate probe runs on the *full item-coverage* sets ($K = 95$ –105 country-level units per item), where the reliability gate is feasible (the floor $\sqrt{2/(K-1)}$ sits below $\tau_D = 0.147$; four of the five items measure $D = 0.139$ –0.142 and pass, while the importance-of-democracy item sits at $D = 0.1473$, a hair’s breadth over the frozen threshold). The *certification* samples of Section 8—countries with ≥ 2 waves, $K = 59$ –77 per item—sit below $K = 94$, where the floor (0.16–0.19) closes the gate outright. And in either denominator the WVS ships no cluster identifiers, so its weights-only design noise gives $\hat{\rho}_{\text{LCB}} \leq 0.09$ —itself biased downward by the missing clustering—and the need gate fails decisively. The survey large enough to *estimate* the deconvolution has almost no design noise to remove; the survey with real clustered design noise has far too few countries to estimate it. The two barriers are anti-correlated, and neither of the two largest repeated cross-national surveys clears both.

The barriers are not survey artifacts. Both gates are distribution-free, so the boundary transfers to any conformal procedure calibrated on estimated objects (Remark 1); a non-survey MRP-style simulation confirms the reliability floor and identifies the one many-area setting, with appreciable posterior uncertainty, where the *need* gate can fire (Supplementary Material).

The AmericasBarometer’s merged design metadata. Its full stratified-PSU structure ships in one merged file (the ESS’s equivalent lives in per-round sample-design files, integrated only from round 9—a data-handling constraint on our window, not an ESS limitation), letting us compute a proper design bootstrap, and the design effect is real (median ≈ 1.13 – 1.20 , up to 1.9 in the most clustered country-years). Yet certification is coarse-robust: the design effect moves band *width* but rarely flips the binary decline decision, because the decision is dominated by the between-country transport spread, not the within-country design noise—the same low- ρ fact, seen from the certification side (details in the Supplementary Material).

What real data validates, and what it cannot. Real data validates the reduction to PCB, the target-blind selection, the efficiency of PCB over the conservative envelope, the real design effect, and the political reclassification of Section 8. It *cannot* validate the high- ρ deconvolution branch or the finite- K remainder—those regimes do not occur at survey scale and rest on the simulation of Section 6 and the theorems, not on the ESS and AmericasBarometer data.

8 Political reanalysis: how many democracies really declined?

The method’s payoff is a sharp distinction between what marginal analysis appears to show and what a design-aware, simultaneous band certifies. We analyze trust in parlia-

ment (`trstprl`) and satisfaction with democracy (`stfdem`) in ESS rounds 9–11 (fieldwork 2018–2024; the window in which PSU and stratum identifiers ship in the integrated file—sample-design files for earlier rounds exist separately and would extend the trajectory, a revision we flag rather than claim), treating each country’s attitude distribution as the object and asking, at each rung of a guarantee hierarchy, in how many countries a decline survives.

The comparative literature has itself grown skeptical of a continent-wide trust decline: where Foa and Mounk (2016) saw “democratic deconsolidation,” Norris (2017) saw trendless fluctuation, Voeten (2017) saw specification-fragile artifacts, and Wuttke et al. (2022)—analyzing eighteen democracies under preregistration—found endorsement of self-government largely intact; the most comprehensive recent accounting finds no universal secular decline in institutional trust at all (Valgarðsson et al., 2025). This qualitative consensus holds that “trust is falling everywhere” is an over-reading; our contribution is a formal instrument that says so, and the wrong-unit diagnosis of *why* it arises. A mean can fall because a mode hollows out or a tail thickens, with different substantive meanings (DiMaggio et al., 1996); requiring the whole distribution to shift is the stronger, and more honest, claim.

A hierarchy of claims. The claims differ in their unit and are not interchangeable. A *pairwise* claim asks whether some pair of waves shows a distributional shift; an *any-pair* claim aggregates these within a country; a *net* claim asks whether the first-to-last span declined; a *persistent country-wide* claim asks whether the *whole* distribution declined *simultaneously* across the trajectory—a single band that must hold over all wave pairs and all thresholds at once; and a *Bonferroni-across-countries* claim further controls the number of countries examined. Each rung is strictly stronger than the last, and conflating them is exactly the over-detection this paper is about.

Which instrument certifies what. Two distinct instruments appear in this section, and they carry different guarantees. The *certification* of a within-country decline is a design-

based simultaneous (sup- t) band on wave-pair CDF differences, built from the stratified-PSU bootstrap (Rao and Wu, 1988; Wolter, 2007)—a design-consistency (asymptotic) guarantee in which the country effect cancels and country exchangeability plays no role. The *transport* band of Theorems 3–5—the finite-sample-exact conformal object—calibrates cross-country prediction and drives the simulation results and the regime diagnostics of Section 7. Headline counts below are certification counts; we say so once here so that the finite-sample language of the theory sections is not silently transferred to them.

The counts collapse up the hierarchy. For `trstpr1` over ESS rounds 9–11 (fieldwork 2018–2024), a marginal pairwise reading flags 20 of 30 countries; requiring an any-pair design-aware band drops this to 12; the net first-to-last decline holds in 6 (Austria, Belgium, Estonia, the United Kingdom, Greece, the Netherlands); and the *persistent, design-aware simultaneous* band over the preregistered low-trust core of the scale—the strongest rung—is certified in **one** country, Greece, which also survives the Bonferroni correction across countries. One disclosure is essential to reading that count: *Greece fielded only rounds 10 and 11 in this window*, so its trajectory is a single wave pair and the persistent, net, and any-pair rungs coincide for it—the top rung degenerates, for the sole surviving country, to the strongest available version of a one-pair claim (the same holds for Czechia and Israel, which do not certify). Among the 27 countries with two pairs, *none* certifies persistence, while the United Kingdom, Belgium, and the Netherlands pass the *plug-in* persistent rung and are demoted by the design-aware layer alone. The honest co-headline is therefore: **net decline in six of thirty; persistent decline in one, and that one on a single pair.** The severity analysis of Section 6 explains why the net rung must carry the secular-decline question: at ESS-like noise the persistent rung has 80% power only against declines of 0.06–0.08 CDF points per pair—crisis-scale collapses—so its non-certifications are weak evidence of absence, while a certification that does pass it is correspondingly strong. For `stfдем` the same pattern holds (23 marginal, 14 any-pair, 8 net, one persistent, again Greece—again on

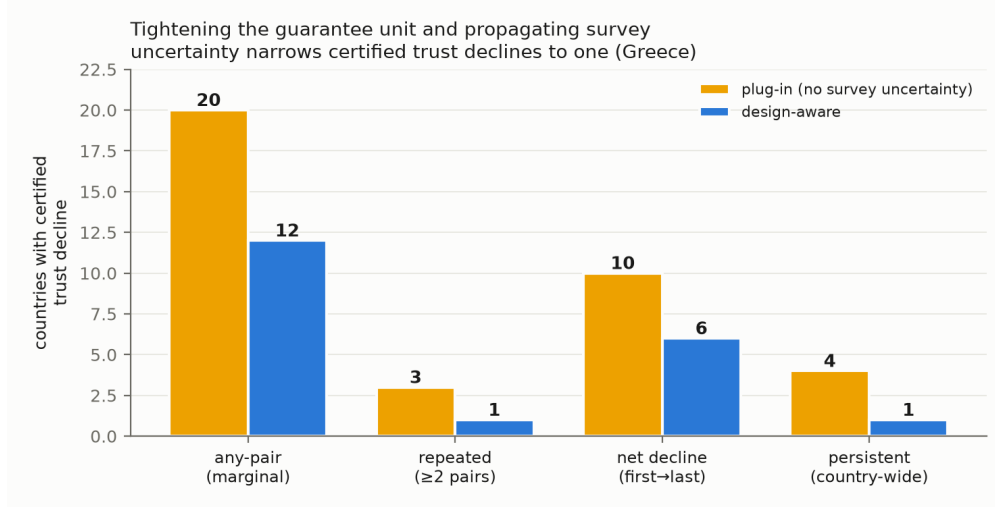


Figure 4: The count of certified declining countries collapses up the guarantee hierarchy (pairwise $\rightarrow$ any-pair $\rightarrow$ net $\rightarrow$ persistent country-wide design-aware), from about twenty marginally to one (Greece).

one pair), though there the Greek decline does *not* survive the additional Bonferroni rung; the across-countries correction is passed for trust in parliament only. Greece is the case the substantive literature independently regards as a crisis-era collapse of legitimacy (Armingeon and Guthmann, 2014; Torcal, 2014), though we note those accounts concern 2009–2015 and the certified pair here is 2020/21 $\rightarrow$ 2023/24, a later episode. The headline is that a *persistent distribution-wide* decline—the strong claim casual readings imply—is certifiable in a single country over this window (Figure 4).

Which countries are reclassified, and the honest word for it. The reclassified countries (Figure 5) are the weak-signal, higher-noise cases, where the apparent trend lies within the design uncertainty the survey warrants. We describe them as *certified by the marginal plug-in but inconclusive under the design-aware simultaneous band*—never as “false positives,” a term we reserve for the simulation where ground truth is known. Because understated design variance would make *more* countries certify, the certified counts err on the inclusive side (the PSU bootstrap resamples m -of- m without rescaling, a known downward variance bias; Section 9), which protects the collapse-of-counts reading though not any indi-

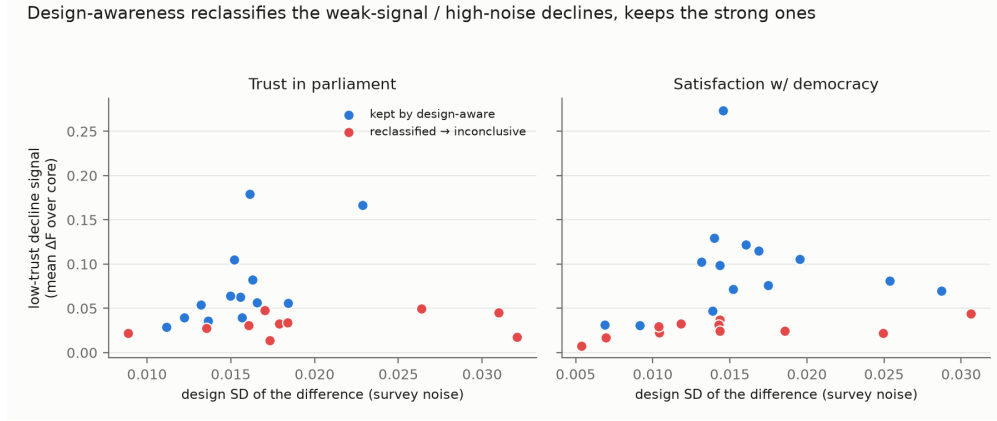


Figure 5: Countries certified by the marginal plug-in but inconclusive under the design-aware simultaneous band are the weak-signal, higher-noise cases; the apparent trend lies within the design uncertainty the survey warrants.

vidual certification.

Two confounds this window cannot exclude. First, round-10 fieldwork coincided with the pandemic: institutional trust exhibited documented rally-round-the-flag movements in 2020–21, so “persistent decline through this window” is a claim made against a rally, and non-certification of two-pair countries partly reflects that. Second, several countries switched to self-completion modes at round 10, and wave-pair CDF shifts across a mode switch confound mode effects with attitude change—a *bias* problem that no design bootstrap absorbs and that our machinery, which corrects sampling variance only, does not address (Section 9 states this scope condition). A mode/fieldwork audit of the certified pairs, and reruns excluding mode-switching pairs, require the microdata and are flagged for the archived replication rerun. Both confounds cut against strong readings in either direction, which is one more reason the net rung, not the persistent rung, should carry the substantive weight of this window.

Robustness across age groups. A preregistered age-group analysis (Appendix B) confirms the “only Greece” result is not an artifact of pooling ages—it holds within the older (50+) and middle (30–49) groups—while youth (18–29) certify zero persistent declines as

an underpowered null rather than youth stability, and emphatically not a story of youth-led democratic disaffection. The full age-band results are in the Supplementary Material.

A second, global target: Foa–Mounk deconsolidation. To show the correction is not a European artifact, we preregistered a reanalysis of the Foa–Mounk “democratic deconsolidation” thesis (Foa and Mounk, 2016, 2017) on the World Values Survey / EVS trends (1981–2022, 59–77 countries with ≥ 2 waves; per the replication package’s preregistration), recoding the deconsolidation battery (importance of democracy, rejection of strong-leader and army rule, support for a democratic system, confidence in parliament) so that deconsolidation is a persistent decline over the preregistered core of each response scale. The verdict is calibration, not denial. A *marginal wave-pair reading of the same data*—a criterion we construct to instantiate the bottom rung, not one any published author uses; Foa and Mounk’s own empirics are cohort-level comparisons—flags 36–64 countries per item; the persistent, weights-aware object is certified in 6–17 (8–24% of countries), a $2.6\text{--}6.5\times$ gap between the rungs. The design-aware layer demotes a further 17–40% per item ($\approx 30\%$ on average) at the persistent bar. Three caveats govern these counts. The WVS ships no PSU or stratum identifiers, so the correction is a *weights-only* respondent bootstrap that understates design variance under the clustered designs most WVS surveys use; understated variance can only *inflate* certified counts, so the rung gap is, if anything, a lower bound—a design-effect sensitivity rerun (variance $\times 1.5\text{--}2$) is flagged for the archived replication. The per-item country lists carry no across-country familywise control (unlike the ESS hierarchy’s Bonferroni rung); at $\alpha = 0.10$ over 59–77 screened countries a handful of boundary-null certifications per item is expected, which is why we aggregate to multi-item co-certification before interpreting geography. And the certification sets ($K = 59\text{--}77$ per item, the ≥ 2 -wave subsets) sit below the $K \geq 94$ feasibility bound, so every WVS band rides the full- α anchors of Theorem 5; the $K \approx 95\text{--}105$ figures in Section 7’s gate probe describe the full item-coverage sets, a different denominator. The youth “embrace authoritarianism” claim is only half-supported (Supple-

mentary Material). The point is the cross-family robustness of the wrong-unit correction, from the dramatic ($20 \rightarrow 1$, the one on a single pair, for European trust) to the moderate ($2.6\text{--}6.5\times$ globally).

The certified core: where the surviving deconsolidation lives. The instrument does not only subtract; aggregating the per-item certifications yields a positive characterization of where persistent deconsolidation is concentrated (Figure 6; replication experiment E30). Of the 38 countries certified on at least one battery item, 13 form a *certified core* certified on two or more of the five: Albania on four; Bosnia and Herzegovina, Ecuador, and Tunisia on three; and Azerbaijan, Finland, Ghana, Iraq, Lebanon, Rwanda, Switzerland, Trinidad and Tobago, and Uzbekistan on two. Three features of this set bear on the debate, each with a stated caveat. *First, the core is not the consolidated West.* It is dominated by post-communist states of the Balkans, Caucasus, and Central Asia and by the Arab-Spring aftermath (Tunisia, Lebanon, Iraq), with fragile Latin American and African cases alongside—young or partial democracies and post-authoritarian societies, not the mature democracies the deconsolidation thesis (Foa and Mounk, 2016, 2017) was formulated about. This geography partly echoes what the post-communist disillusionment literature anticipated—decline from euphoric transition-era baselines (Mishler and Rose, 1997; Inglehart and Catterberg, 2003)—so the certification is best read as formalizing, with per-country error control, a concentration that literature described qualitatively; and several core members (Rwanda, Uzbekistan, Azerbaijan) are electoral autocracies, where falling stated support is a phenomenon of autocratic legitimation and possible preference falsification, not “deconsolidation” in the consolidated-democracy sense. Consolidated Western democracies enter the *core* exactly twice—Finland and Switzerland, on the strongman/army-rule pair—though the single-item set adds Western entries on other items: Canada and Sweden certify on importance of democracy, and Norway and the United Kingdom on confidence in parliament. The accurate summary is that no consolidated Western democracy certifies on

more than one item, and none certifies on support for a democratic system—consistent with the preregistered finding that endorsement of self-government remains intact (Wuttke et al., 2022)—not that the West is absent. *Second, the Finnish–Swiss pair is flagged, not headlined.* Persistent rising openness to strongman and army rule in two of the most stable democracies contradicts most published accounts, and precisely these battery items carry documented translation and response-option instabilities across waves, and both countries’ trend files interleave EVS and WVS administrations; a distribution-wide shift is exactly what an administration or wording change produces, and our machinery corrects sampling error only (Section 9). We therefore report the pair as a candidate finding conditional on a within-program measurement audit, which requires the microdata. *Third, the celebrated backsliding cases are narrower than their reputation—with a window caveat.* Turkey, the Philippines, and Mexico certify only on the importance-of-democracy item. That item enters the battery only from the mid-2000s, so its persistence window is shorter than the other items’ (which reach back to 1981/1995); “only on imp_dem” therefore mixes substance with differential window length, and the interpretive reading—persistent public downgrading of democracy’s importance without wholesale attitudinal collapse—should be held with corresponding looseness. More generally, wave coverage varies by country and item, so multi-item certification is harder where fewer items and waves are measured; a per-country table of qualifying waves and programs is flagged for the archived replication. The counts are descriptive aggregation of per-item certifications; the regional labels are descriptive groupings, not covariates.

9 Discussion

Uncertainty in claims about repeated cross-national surveys is usually computed for the wrong unit—twice: a wave-pair mean contrast stands in for a persistent distribution-wide trajectory, and an estimated distribution for the latent one it samples. We give a multiplicity-honest simultaneous band for the first and a hard boundary for the second: without the

design-noise law the latent uncertainty is non-identified (Theorem 1), and *with* it the correction is still unreachable at cross-national scale, blocked by two anti-correlated barriers no survey clears at once (Proposition 1). The honest deployed object is therefore the reduction—a clustered band exact at any K (Theorem 3) inside a selector that deconvolves only when safe and otherwise abstains, provably correctly (Theorem 5), confirmed on an independent pre-registered design. The substantive payoff is carried by the two reanalyses; the design-aware layer is a diagnostic that certifies, rather than corrects, at survey scale.

When it reduces to ordinary conformal. On real cross-national attitude data the design noise is small relative to between-country spread, so the method reduces *exactly* to clustered population conformal (Section 7): the machinery is diagnostic, *certifying* that the simple band is the honest one and reporting the ρ that justifies it. The deconvolution regime requires *both* many exchangeable units ($K \geq 94$) and design noise comparable to the between-unit signal—a combination cross-national surveys never provide (Proposition 1) but many-unit clustered settings (subnational small areas, schools, clinics) can. A simulation of exactly that setting (Supplementary Material, E31) confirms the payoff is real, not hypothetical: at $K \approx 220$ –300 small areas with posterior noise around 0.7 of the between-area signal, the frozen pipeline activates deconvolution on up to 98% of draws and delivers a band 0.69–0.74 $\times$ the conservative envelope’s width at valid coverage—the positive-regime frontier the survey-scale impossibility leaves open.

Limitations. Four scope conditions bound what the instrument corrects. *Sampling error only.* The machinery corrects design-based sampling variance; it is silent on measurement non-invariance—item wording and translation changes across waves, survey-program (EVS/WVS) house effects, and the round-10 ESS mode switches—which produce exactly the distribution-wide shifts the bands certify. Where a certified pair straddles such a change (flagged in Section 8), the certification is conditional on measurement constancy that the data cannot establish. *Variance-estimator bias.* The deployed PSU bootstrap resamples

m -of- m without Rao–Wu–Yue rescaling (Rao et al., 1992; Wolter, 2007) and drops singleton strata, biasing design variance downward; the package now ships a rescaled variant, and reruns of the shipped CSVs with it are flagged for the archived replication. *A superpopulation postulate.* The transport band’s exchangeability assumption is over a non-sampled, panel-attriting set of countries (Section 3); the certifications do not use it, but the transport results do. *Metadata.* Without design metadata—as in the World Values Survey, which releases weights but limited cluster identifiers—only a weights-only replication is possible; the impossibility result says no design-valid inference can be recovered from the point estimates alone. And the finite- K remainder bound for the deconvolution branch was calibrated on Gaussian data, so it is conservative but not fully robust under strong dependence, pointing to a richer calibration family or a dependence-aware gate—though, since that branch is provably never reached at cross-national scale (Proposition 1), this bears only on hypothetical large- K , high-noise applications. (The base band’s earlier small- K undercoverage is *resolved*: the deployed unstudentized band is finite-sample exact at any K (Theorem 3) at $\leq 5\%$ width on real data.)

The general lesson. Attaching uncertainty to the right unit changes substantive conclusions. For European trust a marginal reading flags about twenty declining democracies; the persistent, distribution-wide, design-aware object certifies one (Greece). For the Foa–Mounk “deconsolidation” thesis the marginal evidence over-counts the persistent phenomenon by $2.6\text{--}6.5\times$ —real in a specific minority, not the broad erosion claimed, and that minority is itself concentrated in post-communist and Arab-Spring-aftermath states rather than the consolidated West (the certified core of Section 8). The pattern is not confined to surveys: wherever a conformal or prediction procedure is calibrated on estimated objects—small-area model outputs, learned features, meta-analytic effects (Remark 1)—the calibration targets carry their own error, and the claim unit is often coarser than the trend asserted. The right response is not to widen every band reflexively but to state the strongest object the data

support, correct the estimation noise when it can be identified, and abstain honestly when it cannot.

A Proofs and efficiency results

Full statements and proofs of Theorems 1–5 and Proposition 1, the efficiency results (AW-1–AW-4), and the modulation self-inclusion deficit are collected in Section S1 of the online supplementary material. Each theorem is additionally paired with a machine-checked contract test that verifies its guarantee numerically.

B Frozen selector specification

The frozen constants ($\rho_0 = 0.47$, $\widehat{\delta}_{\text{UCB}}(D) = 0.0061 + 0.0943 D$, and the remainder), their derivation on the development grid, and the sealed independent-validation protocol (evaluation grid, seed manifest, holdout results) are provided in the replication package.

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Data and Code Availability

All simulation, theory, and benchmark results are deterministic, reproducing exactly under fixed seeds with no external data, and each theorem is paired with a machine-checked contract test. The ESS, World Values Survey, and AmericasBarometer/LAPOP microdata are licensed to their providers and not redistributed; the replication package documents the exact variables, waves, and retrieval steps. The full package—the `pcb` module, all simulations and contract tests, and the analysis pipeline—will be deposited in the Political Analysis Dataverse (Harvard Dataverse) upon conditional acceptance.

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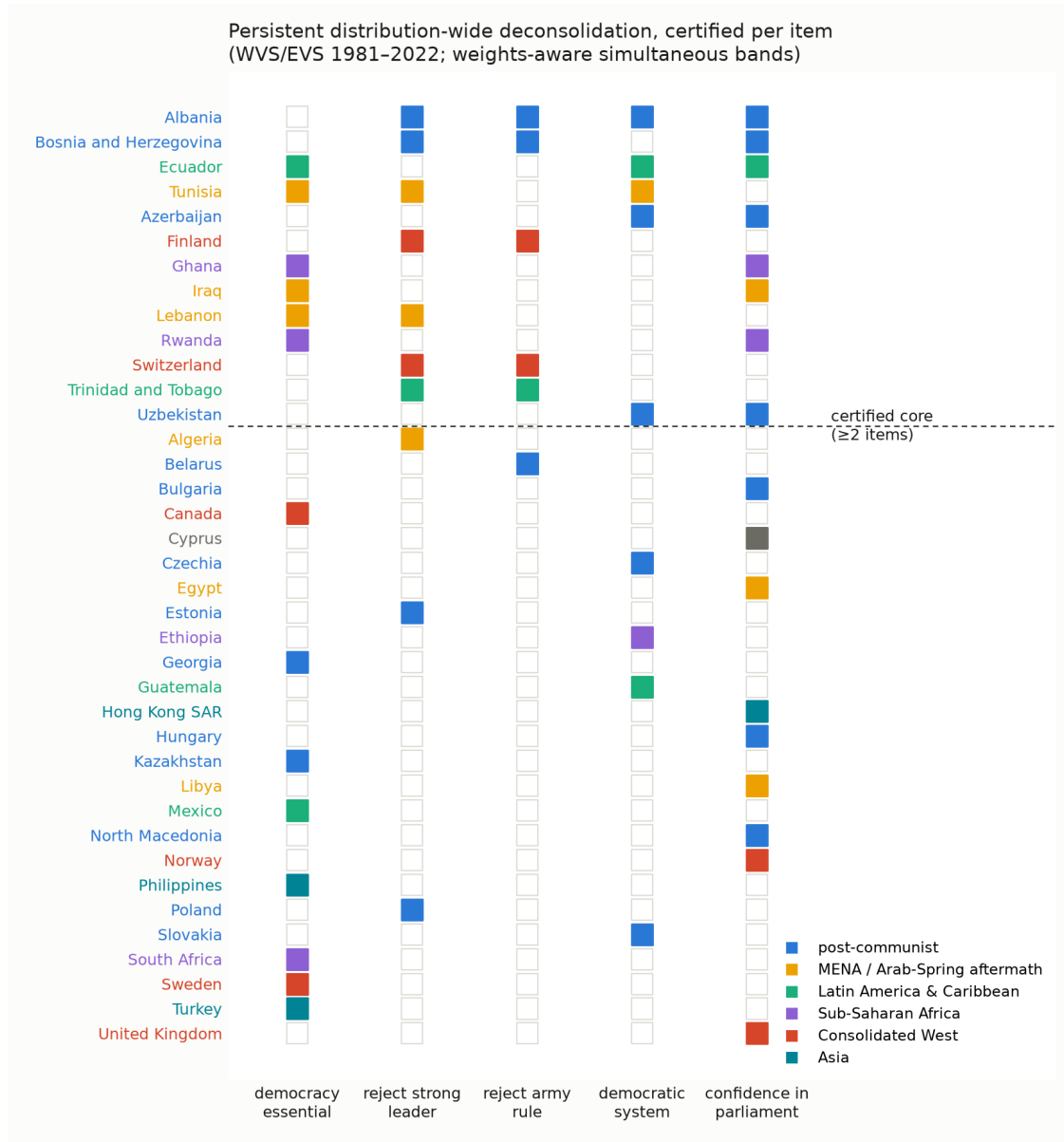


Figure 6: Co-certification across the five deconsolidation battery items (WVS/EVS 1981–2022). Filled cells are country-items where the persistent, weights-aware decline over the preregistered scale core is certified; the 13 countries above the dashed line are certified on ≥ 2 items. The certified core is concentrated in post-communist and Arab-Spring-aftermath states; consolidated Western democracies enter the core only via the strong-man/army pair (Finland, Switzerland), with single-item Western entries on importance of democracy (Canada, Sweden) and parliament confidence (Norway, United Kingdom).